

Quality Agreement and Temperature- Control Protocol

Temperature-Control Protocol for BE-214 Materials

Executed January 15, 2024. Submitted with the complaint on December 2, 2024 in Civil Action No. SYN-WDNC-24-CV-0520.

This quality agreement allocates operational responsibilities between Blue Ember Biologics, LLC and Granite Heron Logistics, Inc. for transportation of development material requiring controlled environmental custody. It supplements Master Logistics Services Agreement BE-GHL-MLS-2023-1101 and applies when a statement of work identifies BE-214 or another accepted program.

The protocol protects data integrity and intended use. It does not transfer Blue Ember's authority to decide whether material meets a release specification, and it does not make Granite Heron responsible for scientific outcomes unrelated to carriage. Instead, it requires the carrier to create conditions and records from which a qualified disposition can be made.

The parties recognize that validation material can become unsuitable for its intended evidentiary purpose even when a later assay detects some biological activity. Consequently, every unauthorized excursion requires documented assessment. Neither automatic destruction nor automatic release follows from a bare alarm. The quality unit must evaluate the actual time-temperature profile, product knowledge, and validation purpose.

1. Quality organizations and decision rights

Blue Ember's quality organization owns specifications, approves transport protocols, reviews deviations, and determines quarantine, release, rework, or destruction. Dr. Mira Solis is the program escalation contact but may delegate technical review to documented qualified personnel. The receiving team may impose an immediate quarantine without awaiting executive approval.

Granite Heron's controlled-transport organization owns carrier training, equipment readiness, dispatch profiles, route records, power-transition procedures, and first response. Adrian Locke provides executive oversight. Tess Holloway manages the Fletcher operation and coordinates an investigation when a shipment experiences an environmental exception.

Solstice Gate Analytics acts as the approved origin for the planned validation lot. It confirms shipper preparation, logger activation, lot identity, seal condition, and origin handoff. Its involvement does not end Granite Heron's responsibility after the driver signs custody.

Quality decisions must be attributable. A salesperson may not approve an excursion; a driver may not declare product safe; and a laboratory observation may not rewrite a custody fact. Each contributor should state the basis and limit of the contributor's role. When teams disagree, they shall preserve the underlying information and escalate without altering source records.

2. Training and qualification

Before handling BE-214, drivers, dispatchers, yard personnel, and supervisors shall complete training on this protocol. Required topics are the frozen range, setpoint, logger identity, alarm response, power transitions, emergency contacts, custody notation, data preservation, and prohibition against unauthorized release statements.

Training includes a practical exercise in switching a trailer from tractor power to electrical standby. The learner must verify that refrigeration continues after the transition rather than assume a connected cable supplies power. Dispatch personnel must demonstrate selection of the active program profile and acknowledgment of test alerts.

Granite Heron will keep the trainee's name, instructor, completion date, materials used, and assessment outcome. Refresher instruction occurs annually and after a material procedure change or relevant deviation. A person whose qualification has lapsed may assist under direct supervision but may not independently complete the critical step.

Blue Ember will brief its receiving personnel on seal verification, quarantine notation, device download, and urgent escalation. Solstice Gate Analytics will train origin staff on the qualified packaging configuration and custody handoff. Each organization remains accountable for its own personnel, including temporary workers and approved contractors.

3. Equipment control and preventive maintenance

Granite Heron shall use equipment capable of maintaining the stated frozen band over the planned route and credible delay. Preventive maintenance must address refrigeration performance, fuel supply, standby connection, controller alarms, door seals, and sensor condition. An overdue critical service removes the trailer from qualified use until disposition by authorized maintenance personnel.

The pretrip check will record equipment identifier, date, inspector, setpoint, observed pull-down, fuel, alarm test, and visible defects. For a planned staging stop, the check also confirms that a compatible electrical source or qualified backup arrangement is available at the intended location.

Calibration of operational sensors must be traceable to an appropriate standard at the interval specified by the equipment program. If later review finds an instrument outside tolerance, Granite Heron shall assess affected shipments and notify Blue Ember when the BE-214 movement may be implicated.

Equipment failure during transit demands protective action, not merely a maintenance ticket. Personnel shall evaluate restoration time, payload temperature, distance to qualified storage, origin return, and destination proximity. Any transfer will preserve seals when possible and create a complete custody entry. Maintenance findings and replaced parts become part of the deviation file when relevant.

4. Logger governance and timekeeping

Blue Ember designates the payload logger in the release or shipment document. For the validation movement, the selected model is CryoTrace CTX-9, expected serial CTX9-44721. Origin staff activate it, confirm battery state, and place it in the qualified monitored location. The driver compares the serial entry before accepting custody.

Device time shall be set to the local civil time identified in the shipment plan, with the applicable offset preserved. For September 12, 2024, entries use Eastern Daylight Time. No user may shift timestamps after collection merely to align an expected event. Any necessary time correction must retain both original and corrected values with an explanation.

The original export is the controlling monitoring record. A portable-document rendering may be generated for review, but calculations and graphs shall remain traceable to the source. Granite Heron shall not trim preconditioning or recovery readings if doing so would conceal context.

An apparent missing interval, implausible jump, or device error triggers a data-integrity inquiry. Other sensors may corroborate or challenge the logger, yet a substitute reading must be identified as such. The parties will not claim machine interpretation of raw data beyond the validated functions documented for the device.

5. Temperature band and response thresholds

Release Specification BE214-RS-03, Revision 3, states the acceptance band for the validation lot: minus eighty to minus sixty degrees Celsius. The trailer operating target is minus seventy. A value exactly at either boundary is within range; a value beyond a boundary is an exception. The specification contains no permitted-duration allowance for this lot.

The first observed exception initiates response even before scientific impact is known. Driver and dispatch must inspect refrigeration status, restore control where possible, record the observation, and contact Blue Ember. A warming trend within the band may justify preventive action when continued rise is reasonably likely.

Recovery is the point at which the designated logger returns to the band and remains there consistently enough to show restored control. Recovery does not erase maximum magnitude or elapsed time. The final summary must identify threshold crossing, peak, intervention, and return times separately.

Overcooling can also affect material and requires the same disciplined process. Personnel shall not change a setpoint to compensate for an earlier warm condition unless directed by qualified technical staff. The goal is safe restoration to the approved band, not mathematical averaging. Product disposition will consider exposure and intended validation use after all available records are collected.

6. Custody entries and staging controls

The origin handoff records lot, shipper count, seals, logger, trailer, driver, date, time, exterior condition, and signatures. Granite Heron's signed acceptance starts controlled custody. Every material stop states arrival, departure, reason, refrigeration mode, and person responsible for any power change.

At Fletcher staging, tractor shutdown shall not occur until electrical standby or another qualified source is engaged and verified on the refrigeration display. The person making the transition records both connection and verification. If verification fails, tractor power remains available while personnel correct the problem or arrange backup.

Keys and access to the sealed load must remain controlled. An opened trailer door requires a stated reason and time. The shipper itself may be opened only by authorized quality personnel unless an immediate safety hazard demands action. Any emergency opening must be witnessed when feasible.

At destination, receiving staff compare identifiers, inspect seals, and note delivery time. A physical-receipt signature is distinct from quality acceptance. If the logger indicates an exception, Blue Ember will mark the lot quarantined and give Granite Heron a receipt copy. The carrier then transmits its route and refrigeration records without waiting for a final rejection decision.

7. Notification and deviation file

For an actual or suspected excursion, Granite Heron shall call the primary Blue Ember quality contact immediately after beginning reasonable protective action. If no connection is made within ten minutes, it shall use the alternate route and document each attempt. Email alone is insufficient for an urgent warming event unless receipt is affirmatively confirmed.

The deviation file begins at first notice and receives a unique carrier identifier. It includes the time series, refrigeration record, route history, custody sheet, alarm messages, interviews, photographs, maintenance findings, and communications. Entries should distinguish personal observation from information learned later.

A preliminary report is due within one business day. A final report should identify root cause, contributing system conditions, product-impact facts, corrective action, responsible owner, and completion date. “Operator error” without examination of training, procedure, interface, and supervision is not a complete root-cause analysis.

Blue Ember may open a linked internal deviation and request reasonable additional information. Its response will state quarantine and eventual disposition. Each company may reserve legal views in a separate communication so the technical record remains usable. Failure to agree on damages must not delay preservation or corrective work.

8. Product-impact and disposition review

Blue Ember evaluates an affected lot through a documented assessment. Relevant information includes starting temperature, boundary crossings, maximum or minimum, duration, packaging qualification, recovery curve, stability knowledge, assay results, intended use, and whether the records satisfy validation expectations.

An assay can inform potency or activity but may not answer every comparability, stability, or documentation question. Conversely, a specification deviation does not permit Blue Ember to disregard reliable evidence that a proposed use remains scientifically supportable. The quality reviewer must explain the relationship between the measured event and the selected disposition.

Possible outcomes are release, conditional use, additional testing, rework, use for a nonvalidation purpose, return, or destruction. For validation lot BE214-VL-2409A, the governing release specification requires compliance throughout custody unless Blue Ember approves a scientifically justified deviation.

Granite Heron receives a disposition summary sufficient to evaluate causation and claim handling, but proprietary formulation detail may be redacted. The carrier may submit technical comments without controlling the outcome. Blue Ember's good-faith quality decision remains subject to ordinary proof if the parties later litigate whether rejection and resulting expense were reasonable consequences.

9. Change control and alternative facilities

A change to origin, destination, packaging, carrier equipment class, temperature band, logger model, manufacturing site, or critical process requires assessment proportionate to risk. The initiating organization states the proposed change, reason, affected records, qualification needed, expected schedule, and approval owner.

An alternative manufacturing facility for BE-214 must show suitable equipment, trained staff, approved materials flow, method transfer, cleaning controls, analytical readiness, and comparability planning. Blue Ember determines whether the site can enter its development program. Commercial capacity alone does not complete quality approval, although quality review should proceed promptly when delay threatens the program.

Expedited change control is permitted. It does not mean undocumented change. Reviewers may accept staged deliverables, conditional approval, focused qualification, or concurrent work when scientific and operational risk remain controlled. Costs and time associated with a reasonable alternative should be recorded so later mitigation analysis uses contemporaneous information.

Neither party may use this section to predetermine whether a specific future offer is reasonable. Capability, deadline, remaining uncertainty, avoided loss, and incremental cost must be considered from the perspective existing when the offer is available. A later outcome cannot substitute for that timely assessment.

10. Audits, records, and retention

Blue Ember may conduct a scheduled audit of Granite Heron’s relevant systems and a focused review after a serious deviation. Granite Heron will provide training, equipment, calibration, maintenance, procedure, shipment, and corrective-action evidence reasonably connected to BE-214. Observations and responses remain together in the audit file; closure of corrective work does not resolve a damages claim.

Shipment and deviation records are retained at least seven years. A reasonably anticipated claim pauses routine deletion for responsive material. Access must protect unrelated customer information and privilege without concealing facts necessary to understand this service. Each organization will identify a custodian able to explain creation, storage, and retrieval.

11. Effective date and approvals

This protocol takes effect January 15, 2024 and continues with the master agreement unless a signed quality amendment supersedes it. A program statement may impose stricter controls but may not silently weaken this protocol. Quality cooperation and legal claim analysis remain distinct: factual disclosure and corrective work do not concede liability, while a reservation of rights does not excuse product protection or record sharing.

For Blue Ember, Dr. Mira Solis approves the protocol as Chief Operating Officer and program quality representative. For Granite Heron, Adrian Locke approves it as Vice President for Controlled Transportation. Tess Holloway acknowledges operational responsibility for implementation at the Fletcher facility.

Blue Ember Biologics, LLC

Dr. Mira Solis — approved January 15, 2024

Granite Heron Logistics, Inc.

Adrian Locke — approved January 15, 2024

Tess Holloway — operational acknowledgment January 15, 2024

The copy appearing in the civil record was attached on December 2, 2024. Its submission date does not change the January 15 execution or effective date.